ICLS – Viscous inclusion experiments

Progress summary of the repository LukasFuchs/ICLS · state of 28 October 2026

1 Motivation and scope

Goal of this first work package: quantify how the stress, strain rate and dissipative energy $\psi = \tau_{II} \cdot \dot{\varepsilon}_{II}$ inside and outside an elliptical viscous inclusion in 2-D depend on the viscosity contrast $\Delta\eta = \eta_c/\eta_m$, the aspect ratio a/b, the orientation θ and the type of background deformation. The results serve as a basis for a possible reactivation criterion for inherited structures in the 3-D regional LaMEM models. The work builds on the earlier FDCSGm (MATLAB) experiments and uses GeoModBox.jl (registered v0.1.1) without modifying it; all additions live in ICLS.

2 Repository and environment

Repository **github.com/LukasFuchs/ICLS** (created with the GitHub CLI, SSH protocol; local clone in WSL under ~/projects/ICLS). Structure:

```
ICLS/
├── README.md, .gitignore (model output excluded: ICLS_jl/**/output/)
└── ICLS_jl/ – Julia environment (Project.toml, Manifest.toml: GeoModBox, Plots)
    └── ViscousInclusion/ – scripts of this work package
```

Scripts run from the terminal (`julia --project=. ViscousInclusion/<script>.jl`) or from VS Code (environment set via `julia.environmentPath`, 4 threads). All scripts locate the environment, the tools and the output via `@__DIR__` and call the tools as `IT.<function>`, so they can be re-run in the same REPL session.

Files in ICLS_jl/ViscousInclusion

File	Content
InclusionTools.jl	Module: analytical solution (EllipseSolution, evaluate, inclusion_values), grid and boundary values, RunInclusionModel (GeoModBox solver), invariants, statistics, error metrics, plotting.
check_analytical.jl	1000 tests of the analytical solution and the closed-form amplification factors (all passed).
ViscousInclusion.jl	Single model: fields and comparison with the analytical solution.
ViscousInclusion_Deta.jl	Sweep over Δη (10^{-5}–10^{5}) for several θ and averaging schemes.
ViscousInclusion_Rot.jl	Sweep over θ for fixed Δη.
resolution_test.jl	Grid convergence (50^2–400^2) for weak and strong inclusions.
compare_averaging.jl	Sensitivity to viscosity averaging and statistics mask.
ParameterMaps.jl	Analytical maps over (Δη, θ, a/b), amplification and shape factors, summary curves; no numerical model.
README.md	Documentation of setup, conventions and outputs.

3 Analytical solution

The solution of Schmid & Podladchikov (2003, eqs. 48–64) for an elliptical inclusion in general shear was implemented in ICLS (Joukowski transform $z = R(\zeta + 1/\zeta)$; arbitrary a/b, θ, Δη, pure and simple shear). It was first verified with an independent Python implementation and then ported to Julia:

- continuity of velocity and traction across the interface; div v = 0; Stokes equations satisfied in the matrix; far-field limit;
- circular limit reproduces GeoModBox's `Dani_Solution_vec!` to ~10^{-10};
- rigid-inclusion pressure (eq. 70) and rotation rate (eq. 65) of the paper reproduced;
- Julia port: 1000/1000 tests in check_analytical.jl passed (incl. 300 general-shear cases of the closed form, section 8).

Inside the inclusion all fields are homogeneous, so the solution gives exact reference values for $\langle\dot{\varepsilon}_{II}\rangle$, $\langle\tau_{II}\rangle$ and $\langle\psi\rangle$. Unlike FDCSGm (circular boundary values with radius a also for ellipses), the elliptical solution now provides consistent boundary velocities and an error reference for ellipses.

The homogeneity of the interior fields is Eshelby's (1957) result, transferred to slow viscous flow by Bilby et al. (1975) for ellipsoids and by Bilby & Kolbuszewski (1977) for elliptical cylinders; Gilormini & Montheillet (1986) give the linear-viscous solution for an elliptical inclusion including the stresses in the matrix, and Fletcher (2009) the extension to anisotropic matrices, Jiang (2013) to power-law materials. Moulas et al. (2014) use the same type of solution for the pressure in and around weak and strong inclusions (shear zones, boudins). Halter et al. (2022) provide a comparable finite-difference code for inclusion–matrix systems.

4 Numerical setup and diagnostics

- GeoModBox staggered-grid Stokes solver (Assembly + LU, defect correction; converges in one iteration). Inclusion via markers (IniTracer2D, 5×5 per cell), viscosity interpolated to centroids and vertices.
- Box 1000 × 1000 m centred on the inclusion; Dirichlet velocities from the analytical solution (bc = :analytical) or from the unperturbed far field (bc = :background).
- Background flow $v_x = \dot{\varepsilon}x + \dot{\gamma}y$, $v_y = -\dot{\varepsilon}y$; pure shear $\dot{\varepsilon} = -\dot{\varepsilon}_{bg}$, simple shear $\dot{\gamma} = 2\dot{\varepsilon}_{bg}$ (same background $\dot{\varepsilon}_{II}$ for both).
- Inclusion statistics on **pure** cells only (no markers of the other phase) and $\tau_{II} = 2\eta\dot{\varepsilon}_{II}$ on the centroids. Cells cut by the interface otherwise dominate the inclusion mean (e.g. +59 % in $\langle\tau_{II}\rangle$ for $\Delta\eta = 10^{-2}$ with arithmetic averaging).
- Error metrics: velocity $\|\Delta v\|/\|v_{ana}\|$ (both components combined); pressure in % of $\tau_{bg} = 2\eta_m\dot{\varepsilon}_{bg}$; each for the whole domain and the bulk (±2 cells from the interface excluded), pressure further split into matrix and inclusion interior.

5 Validation

Viscosity averaging

The three schemes were compared over the full Δη sweep (a/b = 3, 200², four orientations; Figure 1). All of them **overestimate** the inclusion means at every Δη and θ. On the weak side harmonic averaging is best (+0.06 to +1.2 %), geometric lies in between (+0.6 to +2.8 %), arithmetic is worst (+1.4 to +4.8 %). On the strong side arithmetic and geometric averaging saturate at +8 to +10.5 % and +6.4 to +9.4 %, whereas harmonic averaging fails: $\langle\dot{\varepsilon}_{II}\rangle$ stops decreasing and is ~35× too large at $\Delta\eta = 10^5$ (+3500 %; ψ off by $\sim 5\cdot10^6$ %). Bulk velocity errors stay below 0.7 % (geom), 1.4 % (arith) and 0.8 % (harm); the matrix pressure error is ≤ 3.5 %, ≤ 7.1 % and ≤ 2.9 % of τ_{bg}; the pressure inside very stiff inclusions is the least accurate quantity (17–27 % of τ_{bg} for all schemes).

With τ_{II} computed from the stress components instead of $2\eta\dot{\varepsilon}_{II}$ on the centroids, the error moves between quantities: arithmetic averaging then gives $\langle\tau_{II}\rangle$ about 10× too large for very weak inclusions (vertex viscosities of the interface), while harmonic averaging gives a nearly correct $\langle\tau_{II}\rangle$ (+5 %) for stiff inclusions although $\langle\dot{\varepsilon}_{II}\rangle$ is 35× too large. Hence the defaults: centroid τ_{II}, and **:geom** for all sweeps (symmetric in log Δη, never worse than ~10 %). Switching schemes with the sign of Δη was rejected (not possible in dynamic models).

These findings agree with earlier benchmarks of staggered-grid and marker methods: with a viscosity jump inside a cell the scheme drops to first-order accuracy (Duretz et al., 2011), the choice of averaging matters most at strong contrasts (Deubelbeiss & Kaus, 2008), and harmonic averaging gives the smallest velocity and pressure errors for a strong inclusion (Thielmann et al., 2014), as in our bulk errors. New here is that the inclusion means, which the amplification analysis relies on, behave differently: for stiff inclusions harmonic averaging fails for them although the bulk errors are smallest.

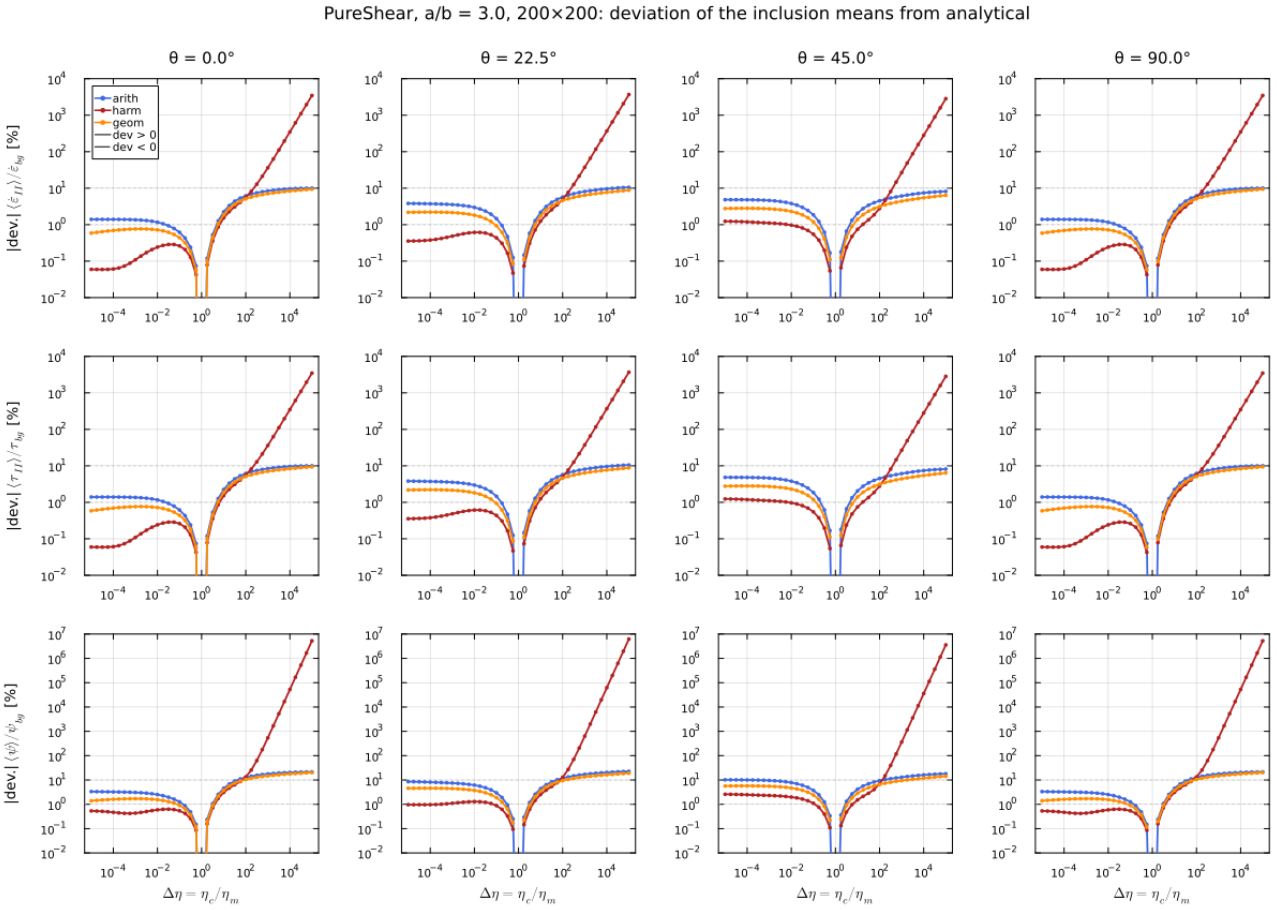


Figure 1. Deviation of the inclusion means from the analytical values for the three averaging schemes (pure shear, a/b = 3, 200², centroid τ_{II}). Log scale of |dev.|; all deviations are positive. Rows $\dot{\varepsilon}_{II}$ and τ_{II} are identical because $\tau_{II} = 2\eta_c\dot{\varepsilon}_{II}$ inside the inclusion.

Resolution

All quantities converge at about first order (geometric averaging, 50² to 400²):

:geom, 400²	circle Δη=10⁻²	circle Δη=10²	ellipse a/b=3 Δη=10⁻²	ellipse a/b=3 Δη=10²
dev. $\langle\dot{\varepsilon}_{II}\rangle$	+0.55 %	+1.74 %	+0.46 %	+2.60 %
L2 v bulk	0.19 %	0.19 %	0.08 %	0.19 %
P matrix [% τ_{bg}]	0.47 %	1.01 %	0.49 %	1.06 %
P inclusion [% τ_{bg}]	0.04 %	5.6 %	0.37 %	7.7 %
obs. order (dev., 200→400)	0.81	0.94	0.65	0.97

Richardson extrapolation ($2f_{400} - f_{200}$) would reduce the inclusion-mean bias to ≤ 0.2 %; not used so far.

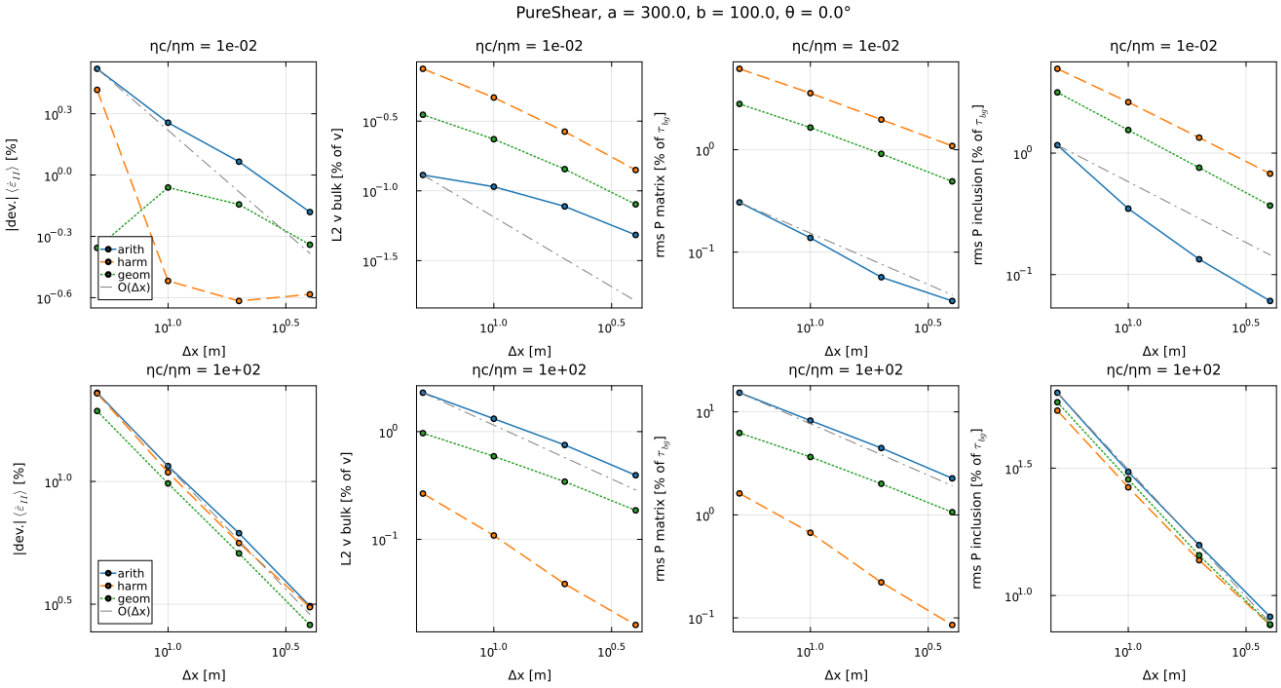


Figure 2. Resolution test for an ellipse (a/b = 3, θ = 0°, pure shear): |dev.| of $\langle \dot{\varepsilon}_{II} \rangle$, bulk velocity error and pressure errors versus grid spacing for three averaging schemes; dash-dotted: O(Δx).

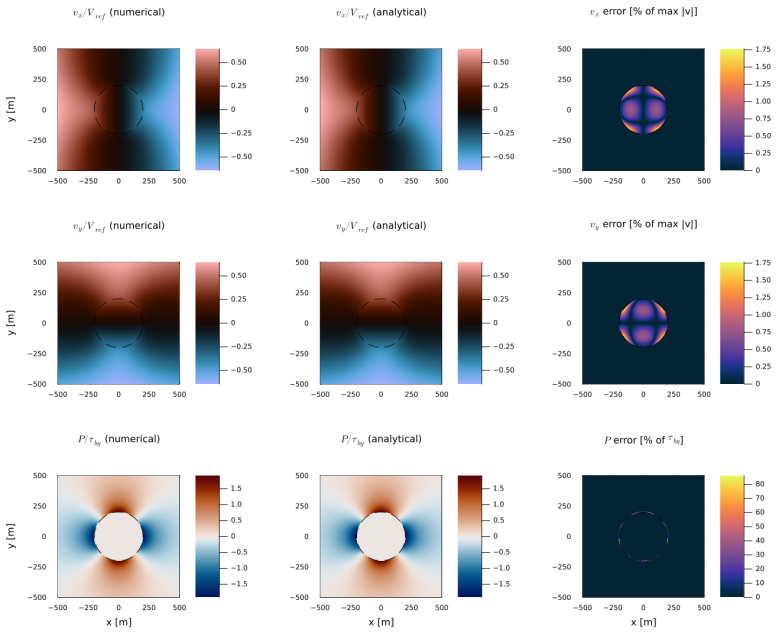


Figure 3. Single model, weak circular inclusion (Δη = 10^{-2}, 200^2, arithmetic averaging): numerical and analytical v_x, v_y, P and errors. The velocity error is largest inside the inclusion (< 2 % of max |v|); the pressure error is confined to the cells cut by the interface.

6 Parameter sweeps (a/b = 3 and 2, :geom)

Δη sweep (a/b = 3, 200^2, θ = 0°, 22.5°, 45°, 90°): numerical and analytical curves coincide over ten decades. Deviation of the inclusion means: +0.6 to +2.8 % on the weak side, growing slowly with log Δη to +6 to +9 % at Δη = 10^5 (more mixed cells behave stiff with geometric averaging). Simple shear (same setup): +3.2 to +4.8 % weak, +5.9 to +8.2 % strong.

θ sweep (a/b = 2, Δη = 10^{-4}, 400^2, θ = 0–45°): the θ dependence is reproduced to ~0.5 %; constant offset of +0.85 to +1.4 %.

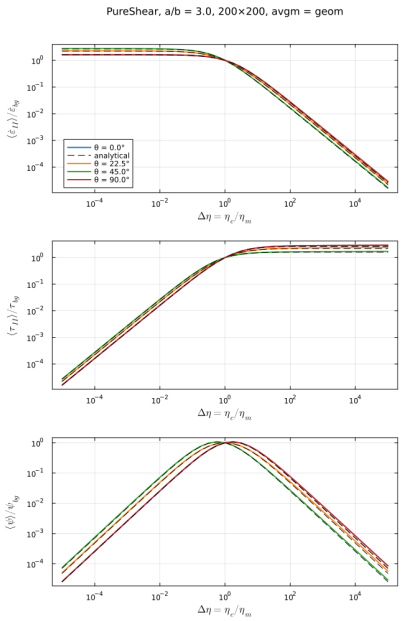


Figure 4. Δη sweep, pure shear, a/b = 3, 200², geometric averaging: inclusion means of $\dot{\varepsilon}_{II}$, τ_{II} and ψ (coloured) and analytical values (dashed).

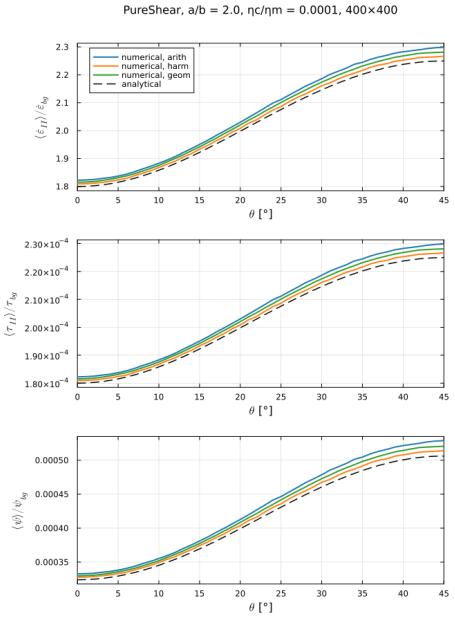


Figure 5. θ sweep, pure shear, a/b = 2, Δη = 10^{-4}, 400², three averaging schemes (θ = 0–45° covers all cases).

Symmetries (exact, used to reduce the parameter space)

- θ and θ + 90° give identical inclusion invariants (90° rotation = reversed shear sense); together with mirror symmetry, θ ∈ [0°, 45°] covers all cases for pure or simple shear alone.
- Simple shear = pure shear rotated by 45° plus a stress-free rigid rotation, so all invariants of the instantaneous problem are identical to pure shear shifted by 45° (confirmed to all digits in the analytical columns). Not valid for finite boxes with background boundary conditions, anisotropy or time evolution.
- Numerically, the error depends on the orientation relative to the grid: diagonal principal strain axes are less accurate (simple shear θ = 0°: +4.5 % vs. pure shear θ = 45°: +2.8 % on the weak side).

7 Analytical parameter maps

ParameterMaps.jl evaluates the inclusion values for a/b = 2, 3, 5, 8, θ = 0–90° and Δη = 10^{-5}–10^{5} (verified against the Python reference to $5 \cdot 10^{-11}$). At fixed Δη, shape and orientation change $\dot{\varepsilon}_{II}$ and τ_{II} by the same factor ($\tau = 2\eta_c\dot{\varepsilon}$) and ψ by its square, so a single shape factor describes the effect.

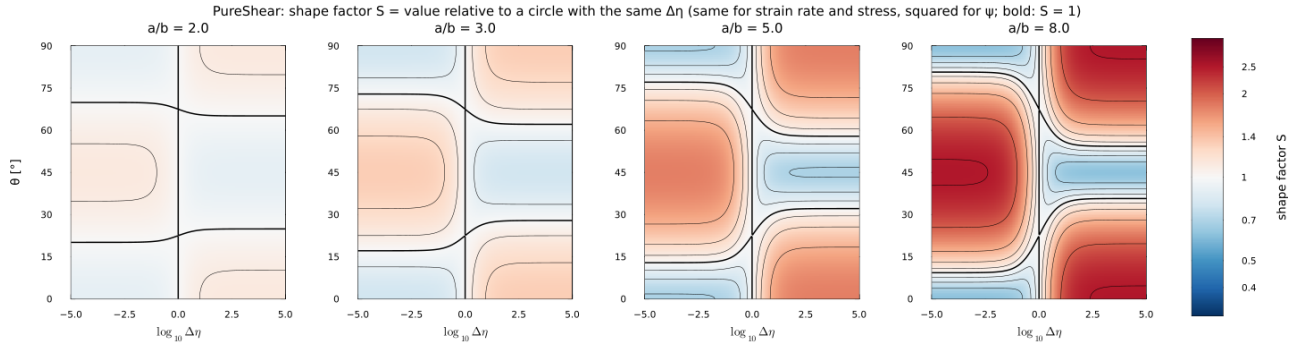


Figure 6. Shape factor S: inclusion value relative to a circular inclusion with the same Δη (bold: S = 1).

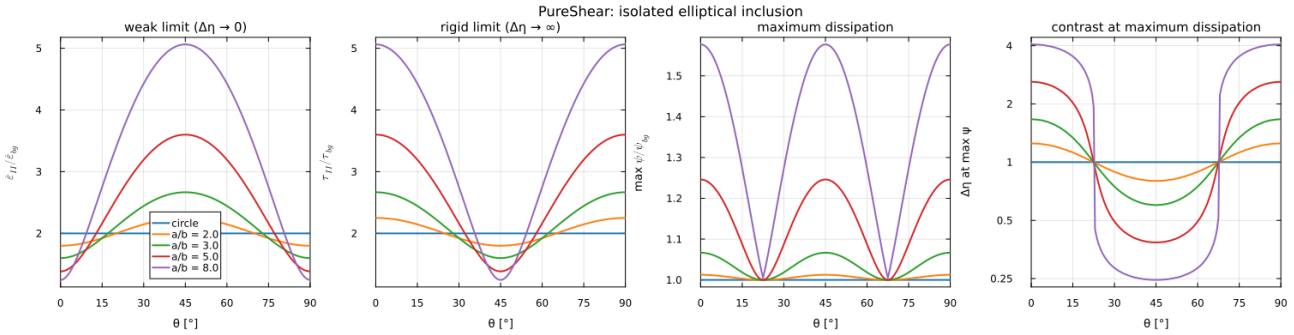


Figure 7. Weak and rigid limits, maximum dissipation, and the contrast at which it occurs, versus θ.

Closed forms (t = a/b, pure shear):

Quantity	θ = 0°	θ = 45°	a/b = 3	a/b = 8
weak limit $\dot{\varepsilon}_{II}/\dot{\varepsilon}_{bg}$	$(t+1)^2/(t^2+1)$	$(t+1)^2/(2t)$	1.60 / 2.67	1.25 / 5.06
rigid limit τ_{II}/τ_{bg}	$(t+1)^2/(2t)$	$(t+1)^2/(t^2+1)$	2.67 / 1.60	5.06 / 1.25
max ψ/ψ_{bg}	$(t+1)^4/(8t(t^2+1))$	same	1.067	1.577
Δη at max ψ	$(t^2+1)/(2t)$	$2t/(t^2+1)$	1.67 / 0.60	4.06 / 0.25

Interpretation

- Elongated weak structures at 45° to the shortening direction concentrate strain most strongly (up to ~5× the background for a/b = 8); aligned with the principal axes they are less effective than a circle.
- Dissipation in the inclusion never exceeds the background by much (≤ 7 % for a/b = 3, ≤ 58 % for a/b = 8) and is maximal at moderate contrasts, not for very weak inclusions: localization of strain and of dissipation are different criteria.
- At θ ≈ 22.5° and 67.5° orientation is neutral; there the dissipation maximum equals the background at Δη = 1.

8 Closed-form amplification factors

From the inclusion coefficient of eq. (55) (with B_1, B_2 from eqs. 57–58 and $\rho^2 = (t+1)/(t-1)$) the amplification factors follow in closed form (equivalent to the inclusion strain-rate ratios of Gilormini & Montheillet, 1986; the compact form and the decomposition below are our formulation). With μ = Δη, t = a/b ≥ 1 and β the angle between the long axis and the background **shortening** direction:

$$A_{\dot{\varepsilon}} = \dot{\varepsilon}_{II,inc}/\dot{\varepsilon}_{bg} = (t+1)^2 \cdot \sqrt{[\ \sin^2(2\beta)/D_1^{\,2} + \cos^2(2\beta)/D_2^{\,2}\ ]}$$

$$D_1 = \mu(t^2+1) + 2t, \quad D_2 = (t^2+1) + 2\mu t, \quad A_\tau = \mu\, A_{\dot{\varepsilon}}, \quad A_\psi = \mu\, A_{\dot{\varepsilon}}^{\,2}$$

Checked against the full solution in 2000 random cases (a/b = 1–20, $\Delta\eta = 10^{-6}$–10^{6}, arbitrary orientation and mix of pure and simple shear): maximum relative deviation $4\cdot10^{-15}$.

- Only the strain-rate part of the background flow enters; the vorticity drops out. Pure shear: β = θ; simple shear: β = θ + 45°. This is the general form of the symmetries in section 6.
- **Decomposition:** $A_{\dot{\varepsilon}} = C(\mu) \cdot S(\mu, t, \beta)$ with the **contrast part** C = 2/(1+μ) (circular inclusion, Eshelby; 2 → 1 → 0 from weak to rigid) and the **shape/orientation part** S, which is the shape factor of Figure 6. S = 1 for a circle or for μ = 1.
- S goes from its weak limit $S_0(t, \beta) = (t+1)^2/2 \cdot \sqrt{[\ \sin^2(2\beta)/(2t)^2 + \cos^2(2\beta)/(t^2+1)^2\ ]}$ through S = 1 at μ = 1 to its rigid limit $S_\infty(t, \beta) = S_0(t, \beta + 45°)$; the shift by 45° explains the mirror symmetry between weak and strong inclusions in the maps. In between, S is not confined to the range of these values: near orientations with $S_0 \approx 1$ it dips below 1 at moderate contrasts (a/b = 8: down to 0.86 at Δη ≈ 0.3 or ≈ 4), so the neutral orientation of the limits is not neutral for all Δη.

- For a weak structure $A_{\dot{\varepsilon}} \approx 2\, S_0(t, \beta)$: the contrast contributes at most a factor 2, the rest comes from elongation and orientation (S_0 up to $(t+1)^2/(4t)$ at $\beta = 45°$, e.g. 2.53 for a/b = 8; down to $(t+1)^2/(2(t^2+1))$ at $\beta = 0°$, e.g. 0.62).
- Threshold criterion $A_{\dot{\varepsilon}}(\mu, t, \beta) = A^*$: explicit for $\beta = 0°$ and 45°, e.g. $\mu^* = [(t+1)^2/A^* - (t^2+1)]/(2t)$ for $\beta = 0°$; otherwise a single root search.
- Implemented as `amplification`, `contrast_factor`, `shapefactor`, `shortening_angle` in InclusionTools.jl and tested in check_analytical.jl (300 cases, general shear).

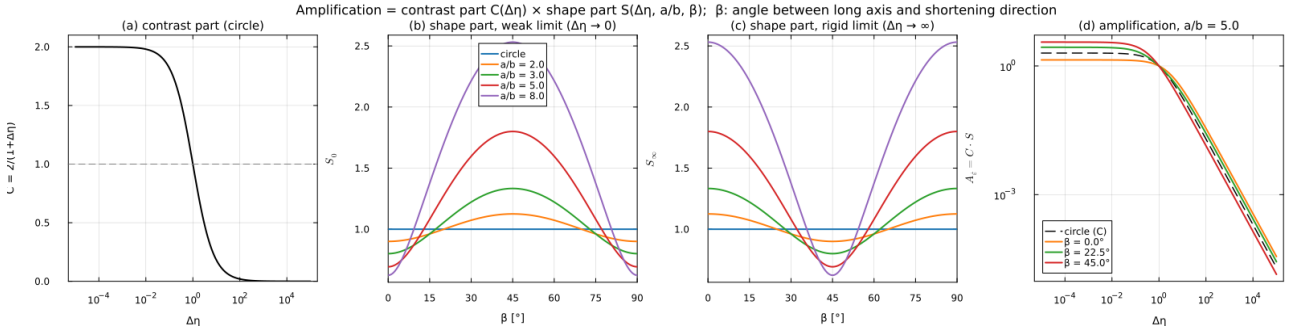


Figure 8. Decomposition of the amplification: (a) contrast part, (b, c) shape part in the weak and rigid limits, (d) $A_{\dot{\varepsilon}} = C{\cdot}S$ for a/b = 5 at three orientations (ParameterMaps.jl, Decomposition.png).

All parameters in one figure

Because $A_{\dot{\varepsilon}}^2$ is linear in $\sin^2 2\beta$, $A_{\dot{\varepsilon}}(\beta)^2 = A_{\dot{\varepsilon}}(0°)^2 \cos^2 2\beta + A_{\dot{\varepsilon}}(45°)^2 \sin^2 2\beta$, with $A_{\dot{\varepsilon}}(0°) = (t+1)^2/D_2$ and $A_{\dot{\varepsilon}}(45°) = (t+1)^2/D_1$. For given Δη and a/b, every orientation therefore lies between the $\beta = 0°$ and $\beta = 45°$ curves, so a single panel shows contrast, aspect ratio and orientation together (Figure 9a). Weak zones: the band widens with a/b and brackets the circle value 2 (a/b = 8: 1.24 at $\beta = 0°$ to 5.06 at $\beta = 45°$); stiff inclusions: the order of the two edges is reversed.

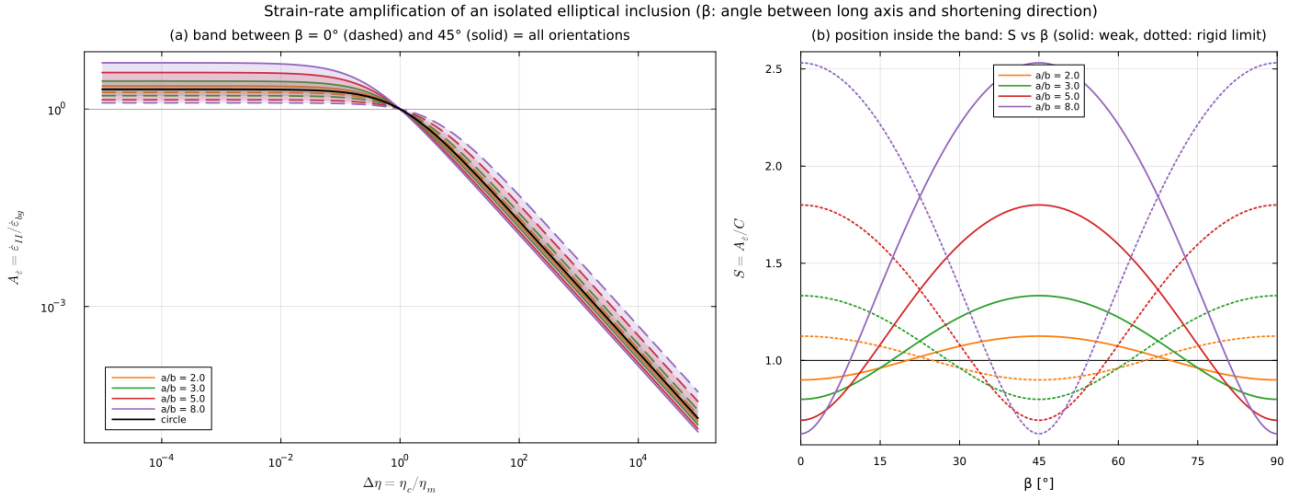


Figure 9. (a) $A_{\dot{\varepsilon}}$ versus Δη; for each a/b the band between $\beta = 0°$ (dashed) and 45° (solid) contains all orientations; black: circle. (b) Where a given β lies inside the band: $S = A_{\dot{\varepsilon}}/C$ versus β in the weak (solid) and rigid (dotted) limits. (ParameterMaps.jl, AmpBand.png).

Maps over orientation and aspect ratio

For a fixed contrast, C is a constant and the pattern over (β, a/b) is set by S alone (Figure 10). For weak zones the strongest amplification is at $\beta = 45°$, and the neutral orientation (S = 1) moves, in the weak limit, from $\beta = 22.5°$ for nearly circular shapes to 17° (a/b = 3), 13° (a/b = 5) and 8° (a/b = 10); closer to the shortening direction an elongated weak zone is less effective than a circle. For stiff inclusions the pattern is mirrored about 45°. Near Δη = 1 the shape hardly matters ($0.82 \le S \le 1.29$ for Δη = 0.5 and 2).

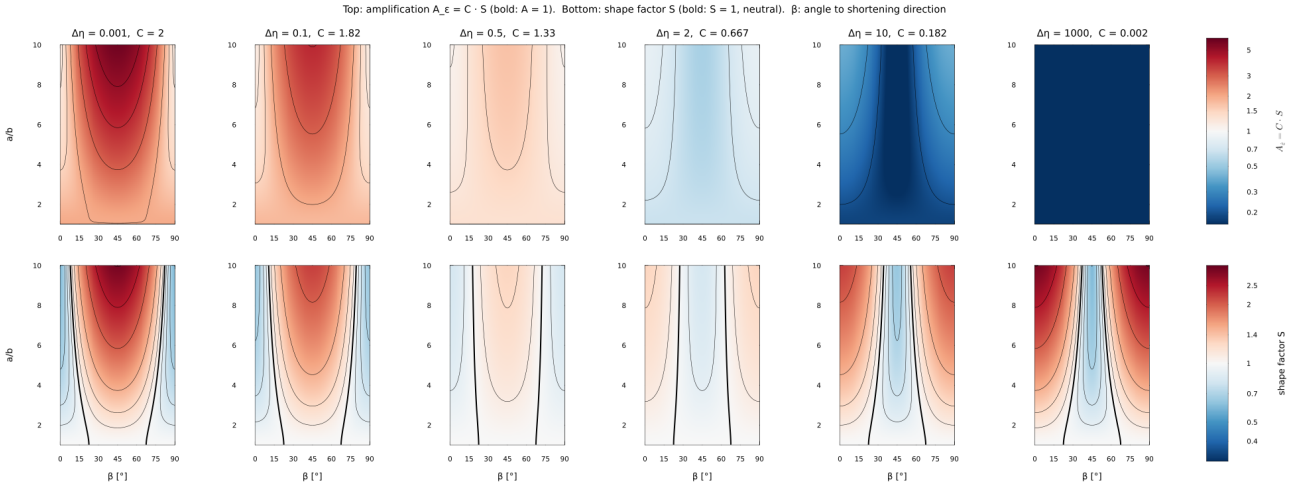


Figure 10. $A_{\dot\varepsilon}$ (top) and S (bottom) over β and a/b for six contrasts; C is given in the panel titles. Thin contours: A = 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 1.1, 1.25, 1.5, 2, 3, 4, 5 and S = 0.4–3; bold: A = 1 and S = 1 (ParameterMaps.jl, AmpShape_ab_beta.png).

9 Where is the amplification strongest?

Values for the range used in the figures ($\Delta\eta = 10^{-5}$–10^{5}, a/b ≤ 8, all orientations), from the closed form; β is the angle between the long axis and the shortening direction.

- **Strain rate.** The strongest amplification, $A_{\dot\varepsilon} = 5.06$ (a/b = 8), is reached for the weakest zone ($\Delta\eta \to 0$), at β = 45° and the largest aspect ratio. There is no optimum inside the range: in the weak limit $A_{\dot\varepsilon} = (t+1)^2/(2t)$, i.e. 2 (circle), 2.67 (a/b = 3), 3.60 (a/b = 5), 5.06 (a/b = 8), growing about as (a/b)/2. The weak limit is approached sooner for compact shapes: 90 % of it at $\Delta\eta \approx 0.11$ (circle), 0.07 (a/b = 3), 0.03 (a/b = 8). Orientation matters as much as shape: at β = 0° the same zone gives only $(t+1)^2/(t^2+1)$, e.g. 1.25 for a/b = 8, four times less than at 45°.
- **Stress.** $A_\tau = \Delta\eta\, A_{\dot\varepsilon}$ has the same maximum value, 5.06 for a/b = 8, but for the opposite case: a rigid inclusion ($\Delta\eta \to \infty$) with its long axis parallel to the shortening direction (β = 0°). A weak zone never concentrates stress: for every $\Delta\eta < 1$, a/b and β, $A_{\dot\varepsilon} > 1$ and $A_\tau < 1$ (checked on a dense grid); the reverse holds for stiff inclusions.
- **Dissipation.** $A_\psi = \Delta\eta\, A_{\dot\varepsilon}^2$ has a maximum at a finite contrast, $(t+1)^4/(8t(t^2+1))$: 1.07 (a/b = 3), 1.25 (a/b = 5), 1.58 (a/b = 8). For a weak zone it occurs at β = 45° and $\Delta\eta = 2t/(t^2+1)$ (0.60, 0.38, 0.25); the same value is reached by a stiff inclusion at β = 0° and the inverse contrast. Very weak zones dissipate little ($A_\psi \to 0$) although their strain rate is largest: the strongest strain localization and the strongest dissipation occur at different contrasts.

Weak zone at β = 45°	**Δη = 0.01**	**Δη = 0.1**	**Δη = 0.3**	**Δη = 0.5**
circle	1.98 / 0.02 / 0.04	1.82 / 0.18 / 0.33	1.54 / 0.46 / 0.71	1.33 / 0.67 / 0.89
a/b = 3	2.62 / 0.03 / 0.07	2.29 / 0.23 / 0.52	1.78 / 0.53 / 0.95	1.45 / 0.73 / 1.06
a/b = 5	3.51 / 0.04 / 0.12	2.86 / 0.29 / 0.82	2.02 / 0.61 / 1.23	1.57 / 0.78 / 1.23
a/b = 8	4.86 / 0.05 / 0.24	3.60 / 0.36 / 1.30	2.28 / 0.68 / 1.56	1.67 / 0.84 / 1.40

Table: $A_{\dot\varepsilon}$ / A_τ / A_ψ for a weak zone in its most favourable orientation. Strain rate: Δη as small as possible; dissipation: $\Delta\eta \approx 0.25$–0.6 depending on a/b; stress: always below the background.

10 Onset of yielding (first step towards a reactivation criterion)

Yield stress as in LaMEM with lithostatic pressure (p_litho_plast = 1): $\tau_y = c\cos\varphi + P_{lith}\sin\varphi$. It depends on depth only, not on the dynamic pressure, so the inclusion pressure (eq. 67) is not needed. Strength ratio $\Lambda = \tau_{y,IWZ}/\tau_{y,m}$ at one representative depth of the zone; zone and matrix isoviscous, background stress $\tau_{bg} = 2\eta_m\dot\varepsilon_{bg}$; onset only (YieldCriterion.jl, yield_onset).

- **Where the stress is highest.** Inside the zone the stress is uniform, $A_\tau\tau_{bg}$. In the matrix the maximum lies on the interface and equals $\max(A_{\dot\varepsilon}, A_\tau)\cdot\tau_{bg}$ (exact: across the interface the tangential strain rate and the shear traction are continuous, so the matrix stress there is

$\sqrt{[(\tau_{tt}/\Delta\eta)^2 + \tau_{nt}^2]}$ of the uniform inclusion stress; along the ellipse the tangent takes all directions, which gives the maximum; confirmed numerically in check_analytical.jl). Interface stress concentrations were analysed by Gilormini & Montheillet (1986) for loading along the inclusion axes (components, for decohesion and matrix cracking; their maximum interface shear traction equals the inclusion stress); the invariant for any orientation and its use for matrix yielding are an extension. Experiments show strain and stress concentrating in the matrix at the tips of weak inclusions (Nardini et al., 2018, 2020; Döhmann et al., 2019). For a weak zone the matrix next to it carries $A_{\dot\varepsilon}\tau_{bg}$, $1/\Delta\eta$ times the stress in the zone, because it has to follow the zone's deformation along the interface with its higher viscosity.

- **Which yields first.** The zone yields at $\tau_{bg}^* = \tau_{y,IWZ}/A_\tau$, the matrix at its edges at $\tau_{bg}^* = \tau_{y,m}/\max(A_{\dot\varepsilon}, A_\tau)$; the smaller value wins. The regime boundary is $\Lambda = \Delta\eta$ for weak zones and $\Lambda = 1$ for stiff ones, independent of shape and orientation (Figure 11a, red line). A weak zone that is weaker in viscosity than in strength ($\Lambda > \Delta\eta$) therefore first makes the matrix at its edges yield.
- **How early.** $\tau_{bg}^*/\tau_{y,m} = \min(\Lambda/\Delta\eta, 1)/A_{\dot\varepsilon}$ for weak zones. Shape and orientation enter only through $A_{\dot\varepsilon}$, so $\beta = 45°$ and elongated zones are the most favourable (Figure 11b; a/b = 8: yielding starts at 28 % of the matrix strength for $\Delta\eta = 0.1$, $\Lambda = 0.6$). The Drucker–Prager zone is isotropic (a weak volume, not a weak plane), so there is no separate frictional optimum orientation as for reactivation of a weak plane (Sibson, 1985). Inherited zones represented as regions of low friction angle in a Drucker–Prager material are used, e.g., by Heron et al. (2016); weak seeds as nucleation sites of shear bands are discussed by Kaus (2010). Closest to the onset criterion: Mandal et al. (2004) show that weak circular (elastic) flaws let the matrix at their tips yield below its yield strength, at a bulk stress decreasing with the competence contrast towards 0.5 for a hole, the same value as our weak-circle limit $1/A_{\dot\varepsilon} = 1/2$; the flaw itself does not yield there, and shape and orientation are not treated; Mandal et al. (2001) use the viscous inclusion stress to predict failure of brittle inclusions. A structured literature review is given in docs/ICLS_LiteratureReview.pdf.
- **LaMEM setup.** Extension in x with $\dot\varepsilon_{yy} = 0$ is pure shear in the x–z section with vertical shortening, so the 2-D theory applies to the cross-section of a zone: a/b = depth extent/width and β = dip angle from the z-axis. For the seven IWZs of the multi-IWZ setup (input cohesions) $\Lambda \approx 0.25$–0.5 at the surface and 0.5–0.8 below ~10 km (Figure 11c). In the current input, min_cohes = 20 MPa raises all IWZ cohesions to the matrix value (unintended; to be checked in the LaMEM work).
- **Limitations.** One depth for the whole zone, no layering, free surface or density contrast, elliptical instead of rectangular cross-section (higher matrix stress at the corners expected), no evolution after the onset.

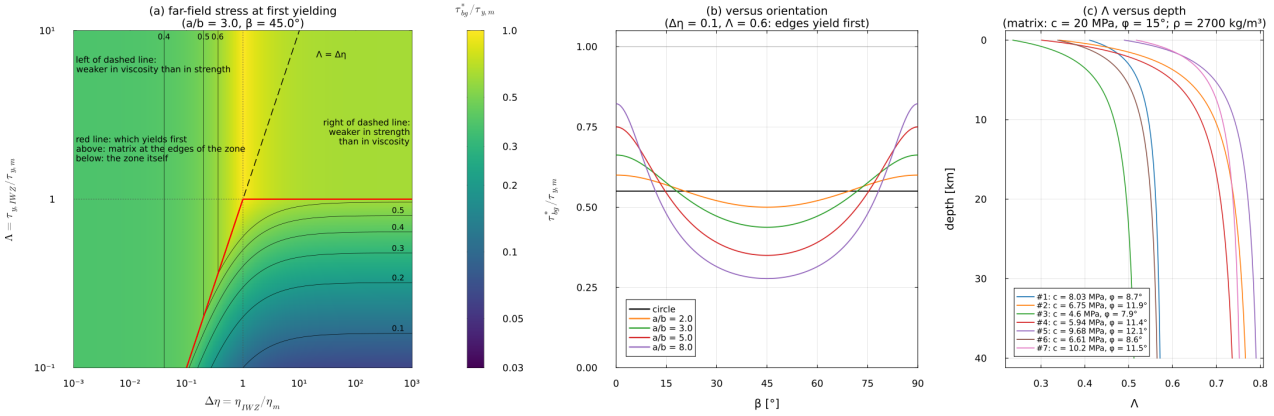


Figure 11. (a) Far-field stress at first yielding relative to the matrix yield stress over Δη and Λ (a/b = 3, β = 45°); red: which yields first, dashed: Λ = Δη. (b) The same versus orientation for several a/b (Δη = 0.1, Λ = 0.6). (c) Strength ratio versus depth for the seven IWZs of the LaMEM multi-IWZ setup (YieldCriterion.jl).

11 Comparison with FDCSGm

- a/b = 3 Δη sweep (PDF summary, p. 5): rigid-limit plateaus of $\langle\tau_{II}\rangle$ roughly 20 % below the analytical values (read off the figure, FDCSGm normalization $\eta_m\dot{\varepsilon}_{bg}$); α = 0° and 90° differ, which cannot occur for an isolated inclusion.
- Rotation in pure shear ($\Delta\eta = 10^{-4}$, a/b = 2, p. 2): same shape (maximum at 45°), but $\langle\dot{\varepsilon}_{II}\rangle \approx$ 2.2–2.65 vs. analytical 1.80–2.25; the new curves are smooth.
- Likely causes: circular-solution boundary values for ellipses and coarse grids (51^2, b ≈ 5 cells); not confirmed in detail.

12 Open points and next steps

- Commit the current state to GitHub.
- **Depth-resolved yield criterion (later).** Yield stress varying over the height of the zone (zone yields first at its top; first yield point on the interface where matrix stress/strength is largest); onset relative to the local and to the surface strength. Then the evolution after the onset (effective viscosity drop, nonlinear).
- Check whether the Eshelby homogeneity can be used for a nonlinear inclusion in a linear matrix (self-consistent effective viscosity); otherwise 2-D visco-plastic inclusion models (GeoModBox or LaMEM) as an intermediate step.
- Configurations without analytical solution: interacting inclusions, inclusion close to box boundaries.
- **2-D GeoModBox models: ellipse vs. rectangle (to do).** The LaMEM IWZs are rectangles in cross-section (a box of given width and depth, rotated by the dip). Compare a rectangular inclusion with an ellipse of the same aspect ratio: inclusion means (expected to be close) and the maximum matrix stress at the edges (expected to be much higher at the corners of a rectangle, so that the matrix at the edges yields earlier than the analytics predict). Square inclusions in pure and simple shear: Treagus & Lan (2003).
- **Transfer to 3-D LaMEM models.** The 2-D amplification $A_{\dot{\varepsilon}} = C{\cdot}S$ is a first-order estimate for an isolated, isoviscous weak zone in an isoviscous medium if the zone is long along strike, the loading is close to plane strain in its cross-section, and it is far from the free surface, boundaries, strong layers and other weak zones.
- **3-D analytical counterpart (to do).** Amplification for ellipsoidal inclusions with finite strike length (Eshelby-type solutions for incompressible viscous ellipsoids: Eshelby, 1957; Bilby et al., 1975; Jiang, 2013, whose strain-rate partitioning tensor is the 3-D analogue of $A_{\dot{\varepsilon}}$), with strike and dip as separate angles. Expected to be lower than in 2-D (e.g. sphere: $5/(3+2\Delta\eta)$, weak limit 5/3, vs. circle: $2/(1+\Delta\eta)$, weak limit 2). More involved than the 2-D case (general Eshelby tensor), but well established.
- **LaMEM baseline sequence (later; LaMEM setup and scripts to be reviewed separately first).** (1) One Newtonian matrix phase and one IWZ phase ($\eta = \Delta\eta{\cdot}\eta_m$), no gravity or equal densities, no free surface, dipping slab through the whole box along y: mean $\dot{\varepsilon}_{II}$ in the zone vs. 2-D $A_{\dot{\varepsilon}}$; box effects estimated first with GeoModBox in the same 2-D section. (2) Actual 3-D geometry with finite strike length. (3) Newtonian layering, then free surface. (4) Plasticity, then the nonlinear flow laws (current setup).
- Role of elasticity (the current multi-IWZ setup is visco-plastic, no shear modulus).
- Key question: how are inherited structures represented in LaMEM – reduced viscosity, reduced friction/cohesion, or both?

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